

Echo Traveller

Description

Echo Traveler is a 2D pixel-art role-playing game (RPG). The player awakens in a mysterious land where everyone speaks music. They need to communicate with non-player characters (NPCs) to uncover the truth behind a vanishing forest. The player makes choices and engages in battles by humming ascending, descending, or stable tones. Every decision branches the story, shaping an ending and a theme song that uniquely reflect each player's chosen paths.

See an example playthrough video here: [EchoTraveller.mp4](#)

Challenges and Changes

Available Unity Assets

There are some free 2D pixel-art assets on the Unity Asset Store, but finding ones that match our theme and remain stylistically consistent across all three scenes—the city under construction, the dungeon, and the mayor's office—was difficult. As a result, we changed the introductory creature from a panda to a deer and purchased a Japanese 2D city asset pack to support the city scene. For the long term, we'll likely need to work with an artist to create more cohesive and theme-appropriate assets.

Audio Input

Using audio input to control gameplay can be tricky for three main reasons.

First, we initially planned for players to learn specific musical motifs to convey attitudes (affirming or denying) or certain keywords, but this requires a level of musical knowledge not everyone has. To make the system more accessible, we simplified it to detect ascending, descending, or stable tones. The trade-off is that this reduces musical richness and immersion in a world where characters “speak” through music.

Second, even with simplified tone detection, noisy environments can still lead to misclassification. We added threshold filters in Chunity to improve accuracy, but audio input still can't match the reliability of traditional keyboard controls. To help prevent errors during dialogue choices, we introduced a confirmation step: once the system highlights the detected choice, players need to press Return to confirm it.

Finally, our playtests showed that not all players feel comfortable using their voice. To address this, we added a fallback option that allows players to navigate dialogue choices with the arrow keys. However, providing an equivalent fallback for the battle system is more challenging without making battles too easy or too difficult.

Collaboration

Creating an RPG involves many different components, so we divided the work based on our strengths and what each of us hoped to learn, then collaborated to bring everything together.

- **Narrative:** Harper and Ling wrote the storyline, including all seven possible branches.
- **Visuals:** After collecting several open-source Unity assets, Ling illustrated scenes 1 and 3, created the player animation, implemented object collisions, arrow-key navigation, and NPC interactions. Harper illustrated scene 2 and several ending backgrounds. Alexey made the start menu.
- **Audio Manager:** Alexey developed the *AudioManager*, a centralised singleton that serves as the game's main audio controller, making it easy to trigger and configure audio events across all scenes. Through *Chunity*, it listens to the player's voice and classifies their humming as ascending, descending, or stable. The audio classification is used for both dialogue-choice navigation and the battle system.
- **Dialogue System:** Harper researched *Ink*, the narrative scripting tool for Unity. Ling built the dialogue UI and programmed dialogue progression with *Ink*, including choice navigation and scene transitions based on player decisions or battle outcomes.
- **Battle System:** Alexey designed and implemented the turn-based battle mechanics, including the instruction page, voice-controlled attacks, the damage/scoring calculations, and Battle Scene's overall layout / "look-and-feel".
- **Music:** Alexey composed the music for scene 1; Harper composed the music for scenes 2 and 3, the battle scene, and all seven ending themes.

Future Plans

We hope to continue developing this project when time allows and eventually release it on platforms like Steam. Our long-term vision is for the current storyline to serve as just one chapter of a larger game, where players can travel across a musical universe and uncover unique stories in each town they visit.